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SUPERSEDING
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21 August 1970

MILITARY SPECIFICATION

CABLES, ELECTRICAL, ALUMINUM FOIL SHIELDED PAIRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

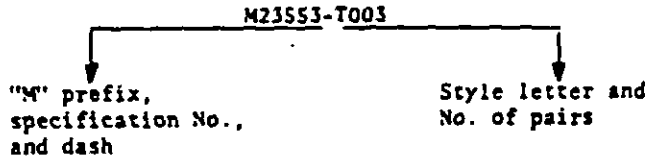
1.1 Scope. This specification covers polyester-backed aluminum shielded pairs of No. 22 American Wire Gage (AWG) wire as audio signal cables for use within buildings and shelters (see 6.1).

1.2 Classification. Cables covered by this specification are classified by the style and number of pairs, as follows:

Style		No. of pairs
S - - - -	Solid conductors, polyvinyl chloride insulated, polyamide jacketed	6, 9, 15, 16, 26, 52, and 104
T - - - -	Stranded conductors, polypropylene insulated	3, 6, 9, 11, 15, 19, and 27

1.2.1 Military part number. The military part number shall consist of the letter "M"; the basic number of this specification; a dash; and a single letter to identify the style, followed by a three-digit number to indicate the number of pairs (see 1.2 and 6.2). NOTE: Where the number of pairs is less than 9, the first 2 digits shall be zeros; where the number of pairs is 11 through 52, the first digit shall be a zero.

EXAMPLE



2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-394 - Plastic Molding Material, (Propylene Plastics, Injection and Extrusion).

MILITARY

MIL-C-17/28 - Cable, Radio Frequency, Coaxial, RG-58C/U.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of.

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MIL-W-16878	-	Wire, Electrical, Insulated, High Temperature.
MIL-W-16878/1	-	Wire, Electrical, Type B, 105°C, 600 Volts (Insulated, High Temperature).
MIL-M-20693	-	Molding Plastic, Polyamide (Nylon), Rigid.
MIL-C-39012/16	-	Connectors, Coaxial, Radio Frequency, (Series BNC (Cabled) - Plug, Pin Contact, Class 2).
MIL-C-39012/21	-	Connectors, Coaxial, Radio Frequency, (Series BNC (Uncabled) - Receptacles, Female, Jam Nut Mounted, Class II).
MIL-I-45208	-	Inspection System Requirements.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-228	-	Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104	-	Limits for Electrical Insulation Color.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF DEFENSE

Handbook H4-1	-	Federal Supply Code for Manufacturers (Part I).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B286-70a	-	Copper Conductors for Use in Hookup Wire for Electronic Equipment.
ASTM D470-71	-	Testing Rubber and Thermoplastic Insulated Wire and Cable.
ASTM D2240-68	-	Methods of Testing for Indentation Hardness of Rubber and Plastics by means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

MUNSELL COLOR CHARTS FOR PLASTIC CABLE AND WIRE

(Application for copies should be addressed to the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, MD 21218.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. First article cables shall be furnished for the inspection specified in 4.3.

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3.2 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the cables to meet the performance requirements of this specification.

3.2.1 Conductor. Conductors for style S cables shall be type I, No. 22 AWG tin coated solid copper in accordance with ASTM B286-70a. Conductors for style T cables shall be type II, No. 22 AWG stranded tin coated copper in accordance with ASTM B286-70a, size designation 22-7.

3.2.2 Plastic. Polyvinyl chloride, polyamide, and polypropylene materials used in any cable shall be certified 100 percent new material which has been through only once the processes essential to its manufacture and application to the conductor or cable.

3.2.2.1 Polyvinyl chloride (PVC). PVC shall conform to type F, grade C of MIL-I-631.

3.2.2.2 Polyamide. Polyamide shall conform to type III, grade E of MIL-M-20693.

3.2.2.3 Polypropylene. Polypropylene shall conform to type IV, grade B, class E of L-P-394.

3.2.3 Tape. Polyester tape shall conform to type G of MIL-I-631.

3.3 Design and construction. The shielded pairs shall be assembled to form a substantially cylindrical core assembled from pairs cabled with various lays to minimize crosstalk. Adjacent layers may be cabled in the same direction or in opposite directions. The 104-paired cable shall be formed in four 26-pair units. The 52-paired cable shall be formed in a single concentric unit. If fillers are used, they shall be of PVC material. The cable length (see 5.1.1) shall be as specified in the contract or purchase order (see 6.2).

3.3.1 Conductors.

3.3.1.1 Drain wire. Drain wire shall be uninsulated conductors as specified in 3.2.1. Each strand shall be free from lumps, kinks, splits, scrapes, and corrosion. Splices made in conductors may be either brazed using a silver alloy solder and a non-acid flux, or welded. Splices shall be butted and shall be free from lumps and sharp projections. In no case shall all strands of a conductor be spliced at one point along the conductor length. The tensile strength of any section of a conductor splice which was brazed or welded, shall be not less than 85 percent of the tensile strength of an adjacent section without a splice.

3.3.1.2 Insulated wire. Insulated wire for style S cables shall be insulated with PVC and jacketed with polyamide and shall meet the requirements of MIL-W-16878 and MIL-W-16878/1, except the conductor shall be as specified in 3.2.1. Insulated wire for style T cables shall be insulated without a jacket with 0.010 $\pm$.002 inch thick extruded polypropylene. The insulated wire in both styles of cable shall be color coded as specified in table I. The first color(s) listed are the color(s) of the helically applied striping. The last color listed is the solid color of the primary insulation of the wire. Helical stripes may be applied to the jacket or primary insulation. Color pigments used shall provide uniform dielectric constant between colors.

3.3.2 Twisting of pairs. The shielded pairs shall consist of two insulated wires and one drain wire twisted together with an average lay not to exceed 6 inches for style S cables or 3 inches for style T cables. The lay shall be determined on a 10 foot length of cable.

3.3.3 Shields. A shield of 0.00035-inch thick aluminum foil cemented to 0.0005-inch thick polyester tape (see 3.3.3.1) shall be spirally wrapped tight around the three conductors with a lay of not more than 4 inches and a minimum overlay of 25 percent in the same direction as the lay of the conductors with the aluminum toward the insulated conductors. Each shield shall be insulated from all other shields. If desired, an additional insulating polyester tape may be spirally wrapped over the shield.

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3.3.3.1 Tape color code. The tape shall be color coded as follows:

First 26 pairs	-	Blue
Second 26 pairs	-	Orange
Third 26 pairs	-	Green
Fourth 26 pairs	-	Brown

3.3.4 Binding tape (104 pair only). For separation on a 104-paired cable, a colored binder shall be applied around each 26 pair units. Colors of the binding shall be readily recognizable. Bindings shall be applied with a lay of not more than 4 inches.

3.3.5 Core covering. The cable core shall be helically wrapped with a single wrap of 0.003-inch thick maximum polyester tape. The core covering shall be applied with an overlay of at least 10 percent.

3.3.6 Marker tape. A tape shall be permanently marked and inserted between the core covering and sheath. The footage shall be sequentially marked every 12 inches. The tape shall be marked at least every 24 inches with the following:

(a) Manufacturer's name, trademark, or code number (the code shall be in accordance with H4-1).
(b) Date of manufacture (within 30 days).
(c) Military part number (see 1.2.1).

3.3.7 Sheath. The sheath shall be of PVC material, slate colored (see 3.4.1), extruded over the core covering or directly over the cabled shielded pairs, as applicable. The thickness shall be as specified in 3.4.5.

3.3.8 Spare pair(s). Spare pair(s) are acceptable for cables of 26 pairs through 104 pairs provided the outside cable diameter does not exceed that specified in 3.4.5.

3.3.9 Patching. The manufacturer's methods and practices of making patches to the conductor insulation and cable sheath shall be subject to approval by the procuring activity. Patching shall be held to a minimum and in no case shall any shipping length contain more than one patch. No buildup or patch that cannot be contained in a single mold shall be made.

3.3.10 Cable put-up. Cable shall be put-up on reels (see 5.1.1).

3.4 Performance.

3.4.1 Color conformance. When tested as specified in 4.5.2, the sheath and insulation color shall correspond to class 1 of MIL-STD-104 and table I.

3.4.2 Hardness. When tested as specified in 4.5.3, the cable sheath shall have a hardness of 80 ±5.

3.4.3 Tensile strength. When tested as specified in 4.5.4, the cable sheath shall have a minimum tensile strength of 2,000 pounds per square inch.

3.4.4 Elongation. When tested as specified in 4.5.5, the cable sheath shall have a minimum elongation of 250 percent.

3.4.5 Sheath thickness and outside cable diameter. When tested as specified in 4.5.6, the sheath thickness and outside cable diameter shall be as specified in table II. The minimum spot thickness of the sheath shall be not less than 70 percent of the specified thickness and the average thickness of any cross section of the sheath shall be not less than 90 percent of the specified thickness.

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TABLE I. Color coding.

Pair number	Color	Pair number	Color
1	Blue and white (or natural)	28	Green-white (stripe) and white (or natural)
2	Orange and white (or natural)	29	Brown-white (stripe) and white (or natural)
3	Green and white (or natural)	30	Slate-white (stripe) and white (or natural)
4	Brown and white (or natural)	31	Blue-white (stripe) and red
5	Slate and white (or natural)	32	Orange-white (stripe) and red
6	Blue and red	33	Green-white (stripe) and red
7	Orange and red	34	Brown-white (stripe) and red
8	Green and red	35	Slate-white (stripe) and red
9	Brown and red	36	Blue-white (stripe) and black
10	Slate and red	37	Orange-white (stripe) and black
11	Blue and black	38	Green-white (stripe) and black
12	Orange and black	39	Brown-white (stripe) and black
13	Green and black	40	Slate-white (stripe) and black
14	Brown and black	41	Blue-white (stripe) and yellow
15	Slate and black	42	Orange-white (stripe) and yellow
16	Blue and yellow	43	Green-white (stripe) and yellow
17	Orange and yellow	44	Brown-white (stripe) and yellow
18	Green and yellow	45	Slate-white (stripe) and yellow
19	Brown and yellow	46 1/	Blue-white (stripe) and violet
20	Slate and yellow	47	Orange-white (stripe) and violet
21	Blue and violet	48	Green-white (stripe) and violet
22	Orange and violet	49	Brown-white (stripe) and violet
23	Green and violet	50	Slate-white (stripe) and violet
24	Brown and violet	51	Red and white (or natural)
25	Slate and violet	52	Red-white (stripe) and white (or natural)
26	Blue-white (stripe) and white (or natural)	Spare (For 52 or 104 pair) 1/	Red and black
27	Orange-white (stripe) and white (or natural)		

1/ At the manufacturer's option, pair number 46 and the "spare" pair may be used to assure the required number of (non-defective) pairs.

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TABLE II. Sheath thickness and outside cable diameter.

No. of pairs	Style	Sheath thickness (nom)	Cable diameter (±.010)
		Inch	Inches
3	T	0.030	0.265
6	S	0.05	0.39
6	T	0.037	0.394
9	S	0.038	0.420
9	T	0.040	0.425
11	T	0.040	0.485
15	S	0.045	0.510
15	T	0.045	0.540
16	S	0.06	0.60
19	T	0.055	0.620
26	S	0.06	0.78
27	T	0.070	0.760
52	S	0.065	0.980
104	S	0.085	1.320

3.4.6 Crosstalk. When tested as specified in 4.5.7, the crosstalk, between two adjacent shielded pairs of the cable with the same direction of lay, shall be not less than 95 decibels (dB) at 70 kilohertz (kHz), 105 dB at 55 kHz, and 120 dB at 40 kHz.

3.4.7 Intercomponent resistance. When measured as specified in 4.5.8, the resistance between the shield and its drain wire shall be less than 1 ohm. The resistance between all other components shall be virtual opens.

3.4.8 Conductor resistance. When measured as specified in 4.5.9, the direct current (dc) resistance of each conductor shall not exceed 16.7 ohms per 1,000-foot length.

3.4.9 Aging stability. When tested as specified in 4.5.10, there shall be no evidence of cracks or flaws in either the insulation material or the sheath.

3.4.10 Dielectric withstanding voltage. When tested as specified in 4.5.11, the cable shall withstand the applied voltage with no insulation breakdown.

3.4.11 Cold bend. When cables are tested as specified in 4.5.12, there shall be no evidence of cracks or flaws in either the insulation or the sheath material, and the dielectric withstanding voltage shall be as specified in 3.4.10.

3.4.12 Dielectric reeling. The cable shall satisfactorily meet the dielectric withstanding voltage of 3.4.10 between the shields after repeated unreeling in accordance with 4.5.13. Any failure within the cable shall be cause for rejection.

3.4.13 Shield coverage. When shielded pairs are tested as specified in 4.5.14, there shall be no exposed areas on the shield.

3.5 Cable identification. In addition to the identification specified in 3.3.6, cables shall be marked with the following:

(a) Each cable length (see 5.1.1) shall be marked on at least one end with its specified length by an appropriate method.
(b) Each end of the cable length containing defective pair(s) shall be tagged. The tags shall indicate specific defective pair(s). For reel marking of length(s) containing defective pair(s), see 5.1.3.

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3.6 Workmanship. Cables shall be processed in such a manner as to be free from all defects which could affect proper functioning in service. The exterior surfaces shall be smooth, uniform, and free from splinters and other defects visible to the naked eye.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection and quality control system. The supplier shall have an inspection system and quality control system in accordance with MIL-I-45208.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection on completed cables (see 4.3).
(b) Quality conformance inspection (see 4.4).
    (1) Quality conformance inspection on individual conductors.
    (2) Quality conformance inspection on completed cables.

4.3 First article inspection. Unless otherwise specified (see 6.3), first article samples shall be fabricated by the manufacturer with tools and methods that, as far as practicable, are the same as those which will be used for quantity production of the product. The manufacture of the cable for a Government contract, prior to approval of the first article sample, shall be at the manufacturer's risk. First article inspection shall be performed on a suitable length of cable. Certification of compliance for the basic material may be obtained from the supplier of the material providing that such certification contains actual tests, examination or other verifiable quality data. First article inspection test reports shall be forwarded to the procuring activity, unless otherwise specified (see 6.2), for approval.

4.3.1 Three first article samples shall be subjected to the tests listed in table III. First article approval for any paired cable shall include approval for any smaller paired cable(s). No failures shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.4.1 and 4.4.2.

4.4.1 Quality conformance inspection on individual conductors. The insulated conductor shall be inspected and shall conform to MIL-W-16878 prior to cabling, except conductor resistance shall be in accordance with 3.4.8 and the insulation material in accordance with 3.2.2. Supplier's test data may be substituted for the inspection.

4.4.1.1 Style S cable conductors. Insulated solid conductors shall be inspected to and conform to MIL-W-16878/1, except for stranding.

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TABLE III. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Visual and dimensional examination	3.2, 3.3, 3.5, and 3.6	4.5.1
Color conformance	3.4.1	4.5.2
Hardness	3.4.2	4.5.3
Tensile strength	3.4.3	4.5.4
Elongation	3.4.4	4.5.5
Sheath thickness and outside cable diameter	3.4.5	4.5.6
Crosstalk	3.4.6	4.5.7
Intercomponent resistance	3.4.7	4.5.8
Conductor resistance	3.4.8	4.5.9
Aging stability	3.4.9	4.5.10
Dielectric withstanding voltage	3.4.10	4.5.11
Cold bend	3.4.11	4.5.12
Dielectric reeling	3.4.12	4.5.13
Shield coverage	3.4.13	4.5.14

4.4.1.2 Style T cable conductors. Insulated stranded conductors shall be tested in accordance with table IV.

TABLE IV. Style T conductor physical and electrical requirements.

Test	MIL-W-16878	
	Condition	Requirement
Insulation tensile strength	A	2400 psi (min)
Insulation elongation	A	200% (min)
Spark test	A	3.4 kV (min)
Dielectric strength	A-2/25 ±5	2.0 kV (min)
Heat resistance	E-96/165	2.0 kV (min) DWV .125 inch (max) shrinkage
Cold bend	E-4/-20	2.0 kV (min) DWV
Soldering	A	.125 inch (max) shrinkage
Flammability	A	30 sec (max) burning time, 7 inch flame travel
Surface resistance	C-96/25/95	1000 megohms (min)

4.4.2 Quality conformance inspection on completed cables. Quality conformance inspection on completed cables shall be in accordance with group A and B tests (see 4.4.3 and 4.4.4).

4.4.2.1 Inspection lot. All cables of the same number of pairs offered for delivery at one time, shall be considered a lot for purposes of sampling and inspection. Quantities of cables which might be delivered against two or more contracts may be combined into one lot for purposes of sampling if produced at the same time.

4.4.2.2 Unit of product. The unit of product for determining lot size shall be the number of reels of cable having the same number of pairs and same approximate total length (500 or 1000 feet).

4.4.2.2.1 Sample. The sample shall consist of that number of units of product required by the sampling plan for the lot size determined in 4.4.2.1.

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4.4.2.2.2 Specimen. A specimen is a length of completed cable from one unit of product.

4.4.3 Group A inspection. Group A inspection shall consist of the examination and tests shown in table V.

TABLE V. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph
Subgroup 1		
Visual and dimensional examination	3.2, 3.3, 3.5, and 3.6	4.5.1
Subgroup 2		
Color conformance	3.4.1	4.5.2
Hardness	3.4.2	4.5.3
Tensile strength	3.4.3	4.5.4
Elongation	3.4.4	4.5.5
Sheath thickness and outside cable diameter	3.4.5	4.5.6
Intercomponent resistance	3.4.7	4.5.8
Conductor resistance	3.4.8	4.5.9
Dielectric withstanding voltage	3.4.10	4.5.11
Dielectric reeling	3.4.12	4.5.13
Shield coverage	3.4.13	4.5.14

4.4.3.1 Sampling for group A, subgroup 1 inspection. A sample shall be selected in accordance with table VI for subgroup 1 inspection of table V. Within each specimen of the sample, the insulated conductors shall be selected in accordance with table VII.

TABLE VI. Sampling for group A, subgroup 1 inspection.

Lot size, number of units (reels) of cable	Sample units to be inspected	Acceptance number	Rejection number
1 to 13	All	-	-
14 to 25	13	0	1
26 to 62	20	1	2
63 to 160	32	1	2
161 to 410	50	2	3

TABLE VII. Sampling for group A, subgroup 2 and measurement of color pair tests.

Pairs in cable	Pairs to be tested
3	3
6	6
9	7
11	7
15, 16	8
19	9
26	10
27	10
52	15
104	20

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4.4.3.2 Sampling for group A, subgroup 2 pair tests. In every length of finished cable, tests shall be performed on the number of pairs specified in table VII. Each of the selected pairs shall be subjected to subgroup 2 tests of table V. If any pair fails in any tests, the entire length of cable shall be rejected. The same pair on the adjacent reels shall be subjected to the same test in which failure occurred.

4.4.3.3 Sampling for measurement of colors. In each specimen (see 4.4.3.1), pairs shall be selected at random in accordance with table VII. The number of pairs on one end of the cable so examined shall be the same as specified in 4.4.3.1.

4.4.4 Group B inspection. Group B inspection shall consist of the test shown in table VIII. Specimens shall be selected in accordance with table IX. For the crosstalk tests, the number of pairs within the specimen shall be selected in accordance with table X and the test results shall be computed in accordance with table VIII to determine acceptance.

TABLE VIII. Group B inspection.

Test	Requirement paragraph	Test paragraph
Crosstalk - - -	3.4.6	4.5.7

TABLE IX. Sampling for group B inspection.

Lot size, number of units (reels) of cable (1000 ft) 1/	Number of specimens for tests required	Maximum specimen failures allowed
1 to 17	3	0
18 to 35	4	0
36 to 70	5	0
71 to 140	7	0
141 to 280	10	1
281 to 560	14	1
561 to 1120	20	2
1121 to 2240	28	3

1/ For 500-foot reels, double the lot size.

TABLE X. Pair sampling per specimen.

Number of pairs in specimen	Number of random pair combinations of crosstalk required per specimen	Maximum failures allowed per specimen 1/
104	20	2
52	15	2
27	10	1
19, 26	10	1
16, 15	8	1
11	6	0
9	5	0
6	3	0
3	2	0

1/ A failure is defined as a pair faulted to one or more adjacent pairs.

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4.4.5 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be inspected using tightened inspection; the sample size shall be the sample shown in table VI and table IX for the next larger lot size for the applicable test.

4.4.6 Inspection of preparation for delivery. The inspection of the preservation packaging, packing and container marking shall be in accordance with the requirements of MIL-C-12000.

4.5 Test procedures.

4.5.1 Visual and dimensional examination. A 3-foot length of cable cut from each end of the specimen shall be examined to verify that the materials, design and construction, cable identification, and workmanship are in accordance with the applicable requirements (see 3.2, 3.3, 3.5, and 3.6). The cable shall be examined for sheath imperfections on all of the surface which is visible without unwinding the cable from the reel.

4.5.2 Color conformance (see 3.4.1). The sheath and insulation color shall be compared with the Munsell color chart for class 1 of MIL-STD-104 and shall match the centroid colors of table XI. If any pair is found not having color as specified, every pair in that specific length shall be compared with the Munsell color card. Also, the reels of cable preceding and following shall be examined to determine the extent of incorrect coloring.

TABLE XI. Munsell centroid colors.

Color	Centroid	Tolerance limits					
		Hue		Value		Chroma	
		H-	H+	V-	V+	C-	C+
Red	2.5R 4/12	10RP 4/12	5R 4/12	2.5R 3/12	2.5R 5/12	2.5R 4/10	None
Orange	2.5YR 6/14	10R 6/14	5YR 6/14	2.5YR 5/14	2.5YR 7/12	2.5YR 6/12	None
Brown	2.5YR 3.5/6	10R 3.5/6	5YR 3.5/6	2.5YR 3/6	2.5YR 4/6	2.5YR 3.5/4	2.5YR 3.5/8
Yellow	5Y 8.5/12	2.5Y 8.5/12	7.5Y 8.5/12	5Y 8/12	None	5Y 8.5/10	None
Green	2.5G 5/12	10GY 5/10	5G 5/10	2.5G 4/10	2.5G 6/12	2.5G 5/10	None
Blue	2.5PB 4/10	10B 4/10	5PB 4/10	2.5PB 3.5/10	2.5PB 5/10	2.5PB 4/8	None
Violet	2.5P 4/10	10PB 4/10	5P 4/10	2.5P 3/10	2.5P 5/10	2.5P 4/8	None
White	N 9/	Value Tolerances: V- is N 8.5/; no V+ limit. Chromaticity Tolerances: 5R 9/1, 5G 9/0.5, 5YR 9/1, 5B 9/0.5, 5Y 9/1, 5P 9/0.5					
Slate	N 5/	Value Tolerances: V- is N 4.5/; V+ is N 6/. Chromaticity Tolerances: 5R 5/0.5, 5B 5/0.5, 5Y 5/0.5, 5P 5/0.5, 5G 5/0.5					
Black	N 2/	Value Tolerances: No V- limit; V+ is N 3/. Chromaticity Tolerances: 5R 2/1, 5B 2/1, 5Y 2/1, 5P 2/1, 5G 2/1					

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4.5.3 Hardness (see 3.4.2). The cable sheath shall be tested in accordance with ASTM D2240-68, on a shore durometer, type A.

4.5.4 Tensile strength (see 3.4.3). The cable jacket shall be tested in accordance with method 3021 of FED-STD-228 or ASTM D470-71.

4.5.5 Elongation (see 3.4.4). The cable jacket shall be tested in accordance with method 3031 of FED-STD-228 or ASTM D470-71.

4.5.6 Sheath thickness and outside cable diameter (see 3.4.5). The sheath thickness and outside cable diameter shall be measured with a micrometer caliper graduated to read in mils or 0.001 inch and having flat surfaces on both the anvil and end of the spindle, each approximately 0.25 inch in diameter. The surfaces of anvil and spindle shall be parallel to within 0.0001 inch.

4.5.6.1 Cables with filler. The sheath thickness and outside cable diameter shall be measured in accordance with method 1331 of FED-STD-228.

4.5.6.2 Cables without filler. A 2-foot length of cable shall be cut from the end of the specimen. The cable diameter shall be at two positions 12 inches apart along the sample length and three measurements 120 degrees apart around each position. The diameters shall be summed and then divided by 6 to obtain the average diameter. The jacket shall then be cut lengthwise in two places 180 degrees apart to obtain two equal sections. Three measurements of thickness shall be made on each piece 6 inches apart. The average thickness is the sum of these thicknesses divided by 6.

4.5.7 Crosstalk (see 3.4.6). Crosstalk shall be measured by applying a known signal voltage into a shielded pair and the voltage induced into an adjacent pair. Specimens shall be 10 feet in length. The equipment required and equipment set-up shall be as shown in figure 1. Cables shall be tested as follows:

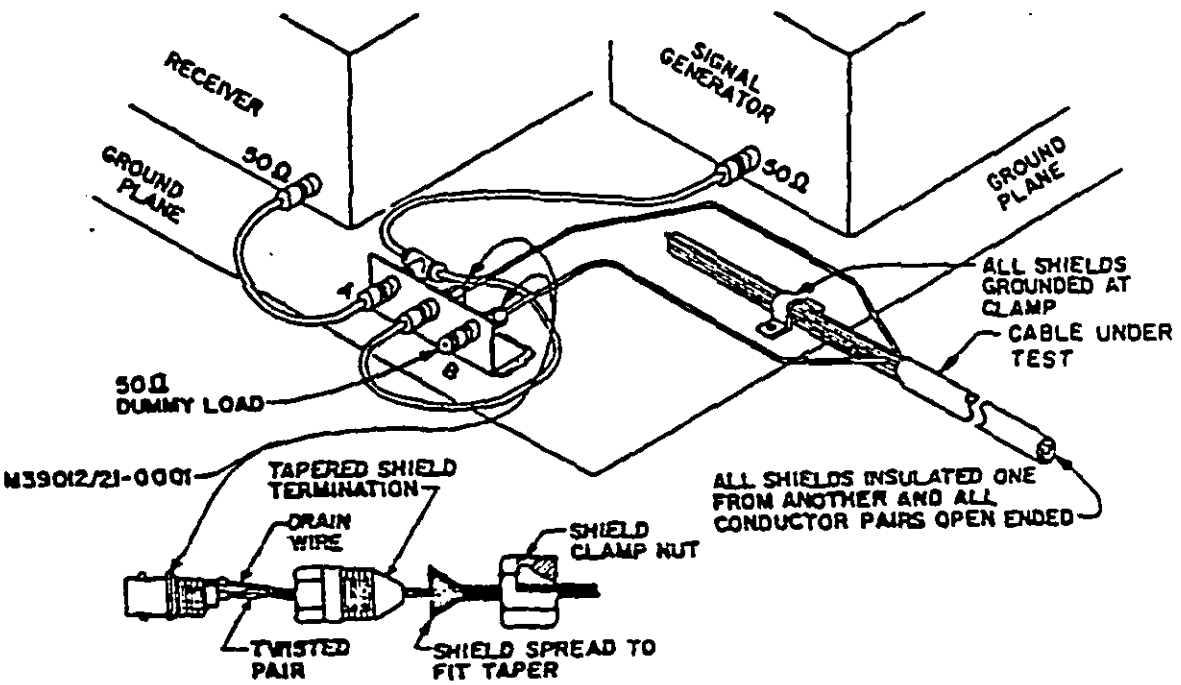
(a) Set the signal generator to a high output level of at least 1 volt, and adjust the input attenuator of the receiver to a reference output reading. This reading shall be recorded and the output level of the signal generator shall remain the same throughout the test.
(b) Remove the receiver and terminate the signal generator with a 50-ohm resistive load.
(c) Connect a shielded pair (which is adjacent to the other shielded pair in the cable under test) to the receiver (see figure 1).
(d) Adjust the receiver attenuator for the same reading as indicated in step (a). The difference in the attenuator readings is the crosstalk (in dB).

4.5.8 Intercomponent resistance (see 3.4.7). Resistance between drain wire and shield, between conductors in a pair, between each conductor and its shield, and between shields shall be measured with an ohmmeter having no more than 1 percent tolerance.

4.5.9 Conductor resistance (see 3.4.8). The resistance of each conductor shall be measured in accordance with method 6021 of FED-STD-228.

4.5.10 Aging stability (see 3.4.9). Unless otherwise specified (see 6.2), the cable shall meet the aging stability test. Specimens of the cable shall be duplicates of those specified in 4.5.12. The specimens shall be suspended in a test oven at the test temperature, without touching one another, or the walls of the oven. Heated air at atmospheric pressure shall be circulated so as to maintain a uniform temperature of 98° ±2°C in the oven. The specimens shall be conditioned for a period of 7 days. After this conditioning period, the specimens shall be removed from the oven and conditioned at room temperature for a minimum of 4 hours. They shall then be subjected to the cold bend test specified in 4.5.12, except that the conditioning and test temperature shall be -35° ±2°C for style S cables and 0° ±2°C for style T cables.

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NOTES:
1. A 50 ohm output signal source from 10 kilohertz to 70 kilohertz is required.
2. Any 50 ohm input receiver capable of measuring to the required limits specified may be used.
3. Other types of terminations may be used in lieu of miniboxes such as coaxial type fittings.
4. A and B shown connected for calibration. Interchange A and B for measurements.
5. All cable assemblies, except cables under test, shall be MIL-C-17/28 (RG-58C/U) with MIL-C-39012/16-0504 connectors.
6. Extreme care shall be used in maintaining continuity between the cable shields and the backshells.
7. Driven line shall be terminated with a 50 ohm load.
8. All shields of the cable under test shall be insulated one from another and all conductor pairs except driven line open ended.

FIGURE 1. Setup for crosstalk test

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4.5.11 Dielectric withstanding voltage (see 3.4.10). Each length of assembled cable shall be tested in accordance with method 6111 of FED-STD-228 and table XII. All connections to the shield of the shielded pairs shall be made to the drain wire, except for the shield-to-shield test.

TABLE XII. Dielectric withstanding voltages for assembled cable.

Points of application	DC or peak voltage		Test duration (min)
	Style S	Style T	
Between conductors - - - - - -	3,600	5,000	3 seconds
Between conductors and shield - - - - - - - - - - -	1,800	2,500	3 seconds
Between shields of adjacent pairs - - - - - - - - - - -	600	600	3 seconds

4.5.12 Cold bend (see 3.4.11).

4.5.12.1 Specimens. For cable(s) with an overall diameter of less than 0.500 inch, the test specified in 4.5.12.2 shall be conducted on three specimens of cable; the length of each specimen shall be at least 150 times the diameter. For 0.500 inch and larger diameter cable(s), the test specified in 4.5.12.2 shall be conducted on two specimens of cable; the length of each specimen shall be at least 120 times the diameter of the cable.

4.5.12.2 Procedure. One end of the test specimen shall be clamped circumferentially at two points, approximately 45 degrees apart, to a mandrel whose diameter is 10 times that of the test specimen. The specimen shall be wrapped around the mandrel for one full turn. The mounted specimen shall then be conditioned for 20 hours at the temperature specified in table XIII. During this conditioning period, the specimens shall be kept reasonably straight. After this conditioning, but while the specimens are still in the cold chamber at the conditioning temperature, the specimens shall be wrapped for three close turns (two close turns for cables having a diameter of 0.500 inch and greater) around the mandrel at a uniform rate of 15 ±3 revolutions per minute, preferably by means of an electrically-driven motor. The cable shall be guided by a free moving sheave or transversing device in intimate contact with the cable at the initial point of bend. The cable shall then be removed from the cold chamber and visually examined except at clamping points. The speciman shall then be tested to 4.5.11.

TABLE XIII. Conditioning temperatures.

Cable diameter (inch)	Style	Temperature
≤.250 - - - - -	S	-50 ±2°C
>.250 - - - - -	S	-40 ±2°C
≤.250 - - - - -	T	-10 ±2°C
>.250 - - - - -	T	0 ±2°C

4.5.13 Dielectric reeling (see 3.4.12). A sample of approximately 500 feet of finished cable selected in accordance with table VII shall be rewound from one reel to another for a total of five times while making at least a 90-degree bend over a

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free rolling pulley or roller having a diameter not to exceed 6 inches. The dielectric withstanding voltage test between all shields of adjacent pairs shall then be performed in accordance with 4.5.11, from each drain wire in turn to all other drain wires.

4.5.14 Shield coverage (see 3.4.13). A length of a foil shielded pair of sufficient length shall be coiled around a mandrel three times the diameter of the shielded pair.

5. PACKAGING

5.1 Preservation-packaging, packing and marking. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified herein, the packaging shall be in accordance with the applicable levels of preservation-packaging, packing and marking of MIL-C-12000 with the following additions and exceptions when level A preservation-packaging is specified:

5.1.1 Level A. Cables shall be furnished on nonreturnable reels which shall contain lengths as specified herein. The 100 feet of cable samples which have satisfactorily passed the crosstalk tests shall be furnished provided that the footage is properly tagged to the cable and marked on the reel. The cable length(s) shall have both ends brought out so that cable can be readily tested without unreeling. The reels shall be constructed with flanges of sufficient thickness to protect the cable from damage during handling, shipment, and storage. The diameter of the reel core shall be not less than 20 times the diameter of the cable.

(a) Put-up for cables having 3 pair through and including 52 pair shall be:
  (1) 85 percent minimum of all cables shall be furnished on reels which contain either 500 or 1000 foot continuous lengths ±10 percent.
  (2) 15 percent maximum of all cables may be furnished in random 100 foot minimum lengths to a maximum of 1100 feet per reel.

(b) Put-up for cable having 104 pair shall be:
  (1) 85 percent minimum of all cable shall be furnished on reels which contain 500 foot continuous lengths ±10 percent.
  (2) 15 percent maximum of all cable may be furnished in random 100 foot minimum lengths to a maximum of 550 feet per reel.

5.1.2 Lagging. Reels shall be closed with wooden lagging. Lagging shall be 2-inch commercial lumber and shall be strapped with a minimum of 2 straps, 1-inch wide. Straps shall be stapled at intervals in accordance with the contractor's commercial practice. A minimum clearance of one cable diameter shall be provided between the inner face of the lagging and cable. The lagging shall be cut even with the outside of the reel flanges nailed with cement-coated steel nails.

5.1.3 Reel marking. In addition to any special marking required by the contract or order (see 6.2), both flanges of each reel shall be marked in accordance with MIL-STD-129, and with the following additional information:

(a) Reel number.
(b) Number of pairs in cable.
(c) Date of manufacture.
(d) Length and location of each piece (in feet) for reels of multiple lengths.
(e) Specific defective pair and cable length (if applicable).

Where practicable, metal tags (in lieu of stencils or paper labels) shall be used and shall be securely attached to the reels. When paper labels are used, they shall conform to the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. These cables are intended to be used as audio communication cables in protective buildings and shelters.

6.1.1 Class S. These cables are intended to be used where flexing is not required and where the dielectric constant is not an important requirement. The temperature range is -55° to +105°C.

6.1.2 Class T. These cables are intended to be used where flexibility or low dielectric constant is required. The temperature range is -10° to +90°C.

6.2 Ordering data. Procurement documents should specify the following.

6.2.1 Procurement requirements:

(a) Title, number, and date of this specification.
(b) Complete part number (see 1.2.1).
(c) Number of first article samples if other than specified in 4.3.
(d) Special marking, if required (see 5.1.3).
(e) Cable length if different from 5.1.1 (see 3.3).
(f) If aging stability test is not required (see 4.5.10).

6.2.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to, the following:

(a) Inspection and quality control system (see 4.1.1).
(b) Test equipment and inspection facilities (see 4.1.2).
(c) First article test reports (see 4.3).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force -
DSA - ES

User activities:
Army - MU
Navy -
Air Force -

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 6145-0678)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. *If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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